

PROJECT REPORT

Linear Relationship Between Media and Sales

Data Analysis and Simple Linear Regression

Prepared by

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This report documents the analysis of the provided Media and Sales dataset and the application of Simple Linear Regression to study their linear relationship.

1. Project Objective

The objective of this project is to analyze the linear relationship between Media and Sales using Python and Simple Linear Regression. The project includes data inspection, visualization, model development, regression-line plotting, model evaluation, and a sample prediction.

2. Dataset Description

The provided dataset contains two variables: **media** and **sales**. Media is used as the feature (independent/input variable), while Sales is used as the target (dependent/output variable). The dataset contains 36 observations and 2 variables.

3. Tools and Libraries Used

Tool / Library	Purpose
Python	Programming language used for the analysis
Google Colab	Environment used to run the Python notebook
Pandas	Dataset loading and data analysis
NumPy	Numerical operations
Matplotlib	Data visualization and regression-line plotting
Seaborn	Data visualization
Scikit-learn	Machine Learning and Linear Regression

4. Methodology

The project follows a sequence of data-analysis and modelling steps. First, the required Python libraries are imported and the dataset is loaded. The data is then inspected for its structure, data types, missing values, and descriptive statistics. Media is selected as X (feature) and Sales as y (target). A scatter plot is used to visually examine the relationship. A Simple Linear Regression model is fitted to the data, predictions are generated, and the regression line is plotted. Finally, R^2 , MAE, MSE, and RMSE are used to evaluate the model.

5. Data Analysis and Visualization

The dataset was inspected using standard Pandas operations including dataset shape, column names, data types, missing-value counts, and descriptive statistics. The scatter plot showed a generally positive relationship: Sales tends to increase as Media increases, although the observations are not tightly concentrated around a single straight line.

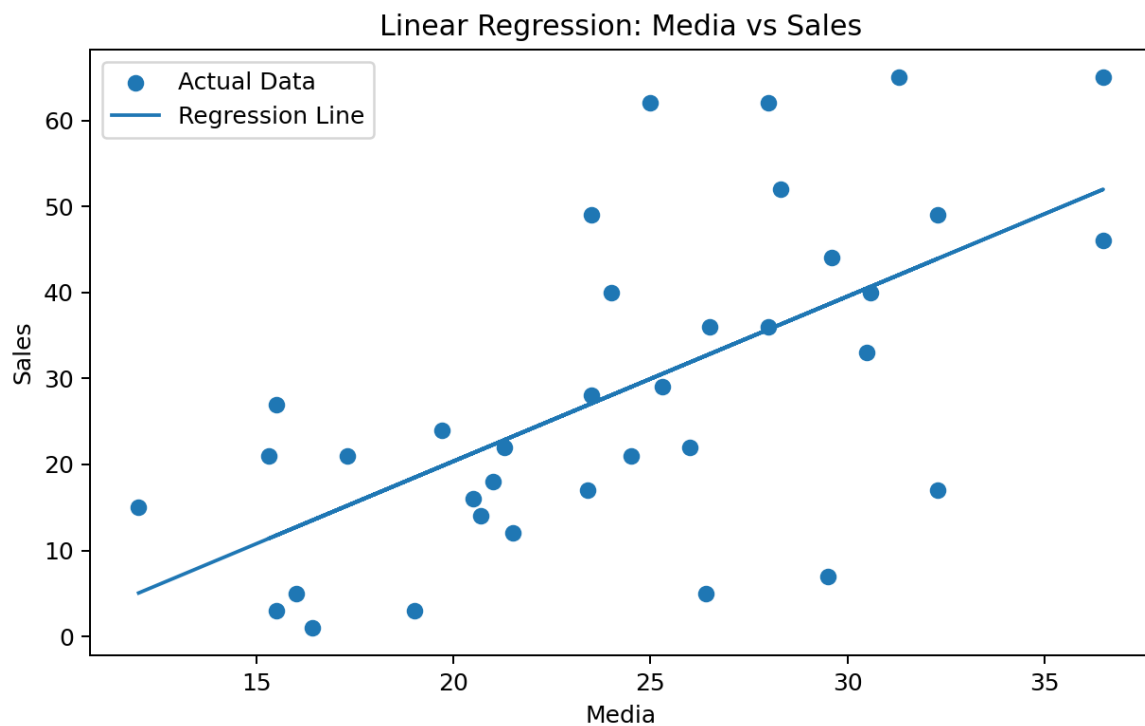


Figure 1: Scatter plot with the fitted Linear Regression line.

6. Feature and Target Variables

Feature (X): Media

Target (y): Sales

The model uses Media as the input variable to estimate Sales.

7. Linear Regression Model

Simple Linear Regression was applied using Scikit-learn. The model learns the best-fitting straight line between Media and Sales by fitting the model to all 36 observations in the dataset.

Regression Equation:

$$\text{Sales} = -17.9208 + (1.9150 \times \text{Media})$$

8. Model Evaluation

Metric	Result	Interpretation
R ²	0.3979	About 39.79% of the variation in Sales is explained by Media in this linear model.
MAE	11.3029	Average absolute prediction error is about 11.30 Sales units.
MSE	206.4141	Average squared prediction error.
RMSE	14.3671	Typical error scale, with greater sensitivity to larger errors.

9. Sample Prediction

A sample prediction was performed by setting Media to 25. Using the trained regression model, the predicted Sales value was **29.9534**.

Media = 25 → Predicted Sales = 29.9534

10. Results and Discussion

The fitted regression coefficient is 1.9150, indicating a positive fitted relationship between Media and Sales. The R^2 value of 0.3979 indicates that approximately 39.79% of the variation in Sales is explained by Media in this linear model. The remaining variation is not explained by this single-feature model. The MAE and RMSE further indicate that prediction errors remain substantial for some observations.

11. Conclusion

The project successfully analyzed the relationship between Media and Sales and applied Simple Linear Regression to the provided dataset. The analysis found a generally positive linear relationship, and the fitted model was used to generate Sales predictions from Media values. For Media = 25, the predicted Sales value was approximately 29.9534. Overall, the project demonstrates the use of data analysis, visualization, and regression modelling to study relationships in a dataset and produce predictions.

12. Project Summary

Item	Value
Dataset variables	Media, Sales
Number of observations	36
Feature	Media
Target	Sales
Regression coefficient	1.9150
Intercept	-17.9208
R^2	0.3979
MAE	11.3029
MSE	206.4141
RMSE	14.3671
Sample Media	25
Sample predicted Sales	29.9534